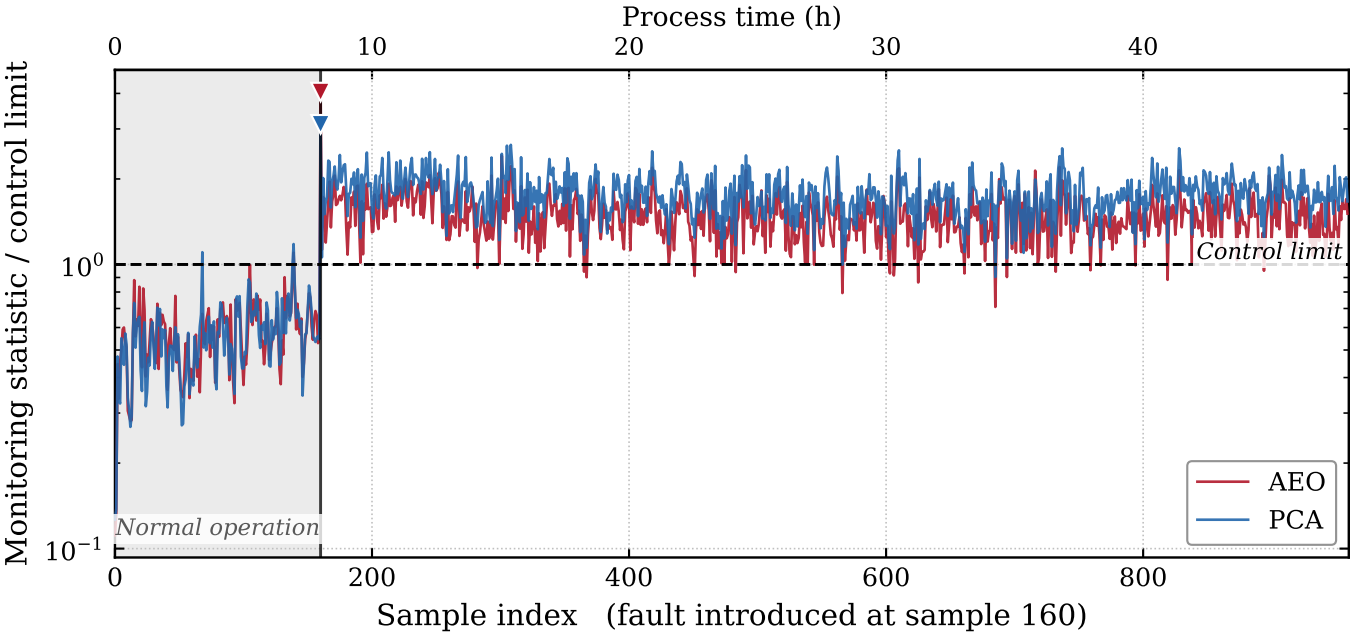
Fault Detection Performance for IDV(4)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	2.3 samples (7 min)	MDR 0.044	detected in 7/7 runs
PCA	delay	0.6 samples (2 min)	MDR 0.002	detected in 7/7 runs

PCA detects 1.7 samples (5 min) earlier; PCA missed-detection rate lower by 96%